

FEM Paradigms for Heterogeneity Modeling in Multi-Material Tissue-Tumor Surgical Twins

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Abstract—Preoperative rehearsal on patient-specific digital twins must predict organ deformation from applied loads and peripheral tissue interactions alone during interactive simulations. Subsurface tumors are invisible to the endoscopic camera, so interpenetration or drift between internal layers cannot be corrected by registration from video-learned surrogates as the simulation advances. Therefore, biomechanical modeling of interactive tissue-tumor digital twins is challenging, as inter-layer displacement continuity requires Finite Element Method (FEM) formulations. We evaluate three FEM paradigms on a single Computed Tomography derived ex vivo liver scene, a) uncoupled and b) constraint-coupled multi-body liver-tumor formulations, and c) a single multi-material volumetric mesh partitioned into Region-of-Interest (ROI) derived multi-elastic subdomains requiring no constraints. Both multi-body formulations lose tissue interface continuity under gravitational settling with Tumor separation from parenchyma, whereas the single-mesh multi-material paradigm preserves continuity by construction. We compare tumor palpation of da Vinci Research Kit (dVRK) patient side manipulator (PSM) tools on the Multi-material FEM model with varying tumor elasticities against a tumor-free baseline. We report a tumor-to-parenchyma stiffness contrast of 1.24 to 1.30 from Lagrange multiplier recovered contact forces, tradeoff between constraint correction fidelity, and interactive frame rates. See Reference Video link¹

Index Terms—Surgical Simulation, Digital Twin, Finite Element Modeling, Robotic Surgery, Tumor palpation

I. INTRODUCTION

Robot-Assisted Laparoscopic (RAL) surgery ensures high quality visual feedback to surgeons while limiting direct tactile perception. This limitation is critical when localizing subsurface tumors for precise margin extraction and subsequent resection planning [1], [2]. Surgeons thus rely on preoperative rehearsals of candidate tumor localization and resection workflows in interactive virtual surgical simulators, where patient specific digital twins are built from pre-surgical medical scans. In an intra-operative surgical setting, augmented reality registration systems solve an inverse problem, where tissue deformations from camera observed surface motions drive the constrain corrections of a biomechanical model overlay [3], [4], on video feeds. However, inter-layer tissue mechanics and its movement dynamics, especially

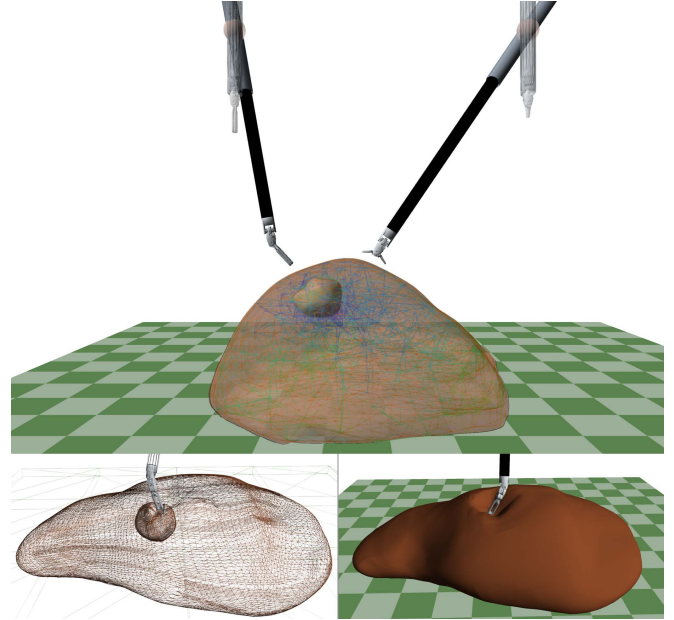


Fig. 1: Liver-Tumor Environment with dVRK PSMs for tumor palpation (top) and visual deformation (bottom) of tumor wireframe (right) and Liver surface (left) for Multi-material FEM configuration.

for subsurface tumors are immune to visible changes from camera observations. Therefore, high fidelity biomechanics formalizations, as a forward problem, are necessary to model physically realistic tissue-tumor digital twins for virtual simulations.

During interactive surgical simulations, an embedded tumor and its surrounding parenchymal matrix must deform cohesively to preserve material continuity along their shared boundary under transient tool loads and gravitational forces. Modeling this multi-material interface at interactive frame rates introduces a classic engineering trade-off between physical realism, numerical stability, and computational speed. While the Finite Element Method (FEM) is the gold standard for enforcing conservation laws to derive volumetric deformation under dynamic loads, conventional interactive pipelines struggle with material heterogeneity. FEM formulations resolve this material heterogeneity either through independent, stacked homogeneous multi-body meshes linked by bonded contact constraints, or through element-wise elasticity variations across a unified grid. Managing steep material stiffness gradients under real-time constraints can introduce numerical artifacts—such as geometric boundary inter-penetrations, mesh drift, or constraint locking leading to simulation instabilities of surgical twins.

To address these limitations, this paper evaluates the

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¹<https://drive.google.com/file/d/193szL0twXVmb6px6PGTqssxt0G4pg4S/view>

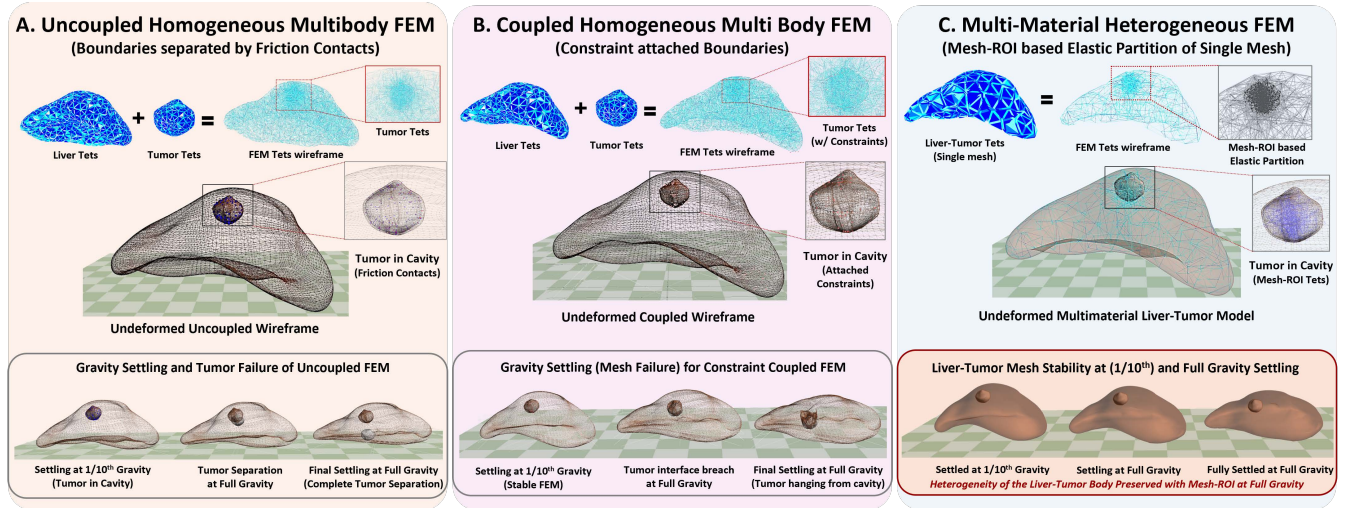


Fig. 2: FEM formulation paradigms for heterogeneity modeling in liver-tumor digital twins. **a) Uncoupled:** Independent tumor and parenchyma tetrahedral meshes interacting via contact friction without explicit coupling constraints. **b) Constraint Coupled:** Independent multibody meshes where proximal cavity and tumor boundary node pairs are linked via structural constraints to couple deformations. **c) Multi-Material:** A unified volumetric mesh partitioned into distinct subdomains using the tumor surface ROI for direct, constraint-free multi-elasticity allocation. Bottom rows illustrate gravity settling: multibody frameworks lose interface continuity, whereas the multi-material framework successfully preserves structural heterogeneity under full gravity.

impact of structural heterogeneity across three distinct FEM paradigms utilizing a Computed Tomography (CT)-derived ex vivo liver scene palpated via a da Vinci Research Kit (dVRK). We compare three FEM formulations: an uncoupled and constraint coupled multibody Liver-tumor FEMs held by friction alone or attachment constraints, and a constraint-free Single volumetric liver-tumor mesh partitioned into Region-of-Interest (ROI)-derived multi-material liver and tumor subdomains, against a tumor-free control baseline. We demonstrate that the multimaterial FEM formulation preserved structural integrity and boundary continuity under palpation and gravitational settling whereas the multibody FEM formulations fail to maintain stability within interactive frame rates.

Although prior surgical robotics research explored control mechanisms [5] for tumor detection via 2D tactile imaging [6], very limited studies [7] addressed rigid-to-multilayered soft body tool-tissue interaction mechanics using Finite Element Analysis [8], [9]. Prior quasi-static FEM indentation simulations [10] on skin-fat-muscle models in ANSYS and Abaqus, showed reduced peripheral strain distribution around subsurface tumor at peak loads under both linear-elastic (5N) and hyperelastic (2.5N) simulations. Experimentally, we reconstructed phantom-embedded rigid tumors to 89% geometric F-Score in under 100 palpation measures using a triaxial force-motion probe on a Franka robot with Bayesian search and impedance-controlled tumor contour tracing [11]. As rigid inclusions in compliant substrate lack anatomical accuracy, characterizing them through computational models in multiphysics simulations serve a cheaper alternative and direct toward anatomical twin building for simulated surgical rehearsals.

II. METHODS

A. Scene construction and Meshing

The liver geometry is derived from Computed Tomography (CT) segmented geometric 3D reconstructions, while an

arbitrary ellipsoidal Tumor geometry is used for experimentation. For multibody FEM modeling a tumor cavity through Boolean subtraction is created in the liver geometry within 1.5cm depth from surface to embed the subsurface tumor. Whereas, for the single mesh multimaterial liver-tumor geometry we embed the tumor points into the liver geometry at the prescribed depth for refined volumetric discretizations using Delaunay-based adaptive tetrahedralization (tet). Meshing is optimized for the Tumor convex hull along the Single mesh to resolve localized stress gradients and coarsened in the parenchyma to limit computational cost. Liver parenchyma is assigned $E = 5.50\text{kPa}$ and $\nu = 0.45$; the tumor is varied with $E = 5.5, 55.0, \&75.0\text{ kPa}$ and $\nu = 0.35$. The mesh resolution is comparable across both multibody (with 1013 and 3371 tumor and Liver tets) and multi-material (comprises 4755 total tets) formulations, respectively.

B. Paradigms for Heterogeneity Formulations

Three formulations represent the same geometry and material assignment.

1) *Uncoupled multi-body:* Liver and tumor are independent volumetric meshes with separate linear corotational FEM force fields and collision models. The geometric boundary inclusion is retained solely by frictional contact within the cavity, with no coupling operator.

2) *Constraint Coupled multi-body:* The same two meshes, each with a linear corotational FEM force field, are joined by bilateral attachment constraints between cavity-surface and tumor-surface node pairs, resolved in the constraint system alongside contact.

3) *Mesh-ROI derived Multi-material FEM:* A single continuous tetrahedral domain $\Omega = \Omega_L \cup \Omega_T$ is partitioned by convex-hull Regions of Interest (ROI), with piecewise-constant elasticity $E(X)$ and Poisson's ratio $\nu(X)$ assigned per subdomain within a corotational elastic FEM force field. Displacement continuity holds by construction, as interface nodes are shared and no coupling operator is required.

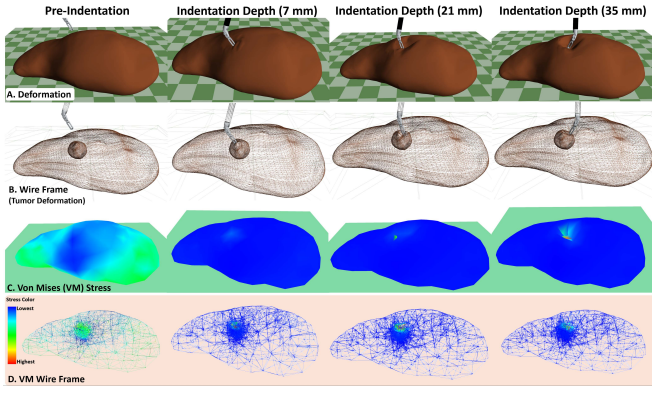


Fig. 3: Visual Deformations and von Mises Stress Results of the Palpated Liver-tumor Heterogeneous Multi-material FEM body at different indentation depths: a) Liver depressions at the indented site, b) Tumor deformation in Liver Wireframe due to indentations at different depths, c) von Mises Stress as visible on the Liver surface due to indentations, d) Stress visualized at each FEM nodes with red (Highest) to blue (Lowest) color map distribution.

C. Robotic Actuation and Physical Measurements

Each FEM formulation is first settled under gravity (g), evaluated at $0.1(g)$ and $1.0(g)$. Surviving formulations are then palpated by a kinematically modeled dVRK PSM Cadiere Forcep following an identical scripted trajectory, actuated through differential-Jacobian inverse kinematics under a Remote Center of Motion (RCM) constraint.

Implicit time integration uses backward Euler to resolve the unconstrained free motion of the FEM nodes. The sparse LDL^T direct solver remains stable at $\Delta t = 0.01s$, whereas the conjugate gradient solver requires $0.001s$ as its approximate solution degrades under the tumor-parenchyma elastic contrast. Contacts are enforced under Signorini’s law [12] and resolved by a Projected Gauss-Seidel solver returning Lagrange Multipliers, from which nodal contact forces follow by scaling the constraint directions and dividing by the timestep. This solve requires a compliance matrix, supplied by a constraint correction that also applies the corrective displacement. We evaluate both variants: a generic correction backed by the direct solver, which derives compliance from the assembled system matrix and retains the spatially varying elastic properties, and an uncoupled correction paired with conjugate gradient, which substitutes a scalar diagonal approximation.

Per timestep we log tip pose, indentation depth, resultant contact force recovered from Lagrange multipliers without a simulated force sensor, peak von Mises stress, tumor and liver centroids, and wall-clock step time. Reported metrics are relative tumor centroid displacement, signed interface distance, apparent stiffness contrast at a reference indentation depth varying $13 - 35mm$ below surface.

III. EXPERIMENTS AND RESULTS

The simulation environments are built on the SOFA physics Engine using the “sofaPython3 API” where an Nvidia RTX-4070 GPU is leveraged for seamless scene rendering. Although most of the physics solver and collision pipeline load is processed by a multithreaded 20 core CPU, the OpenGL model uses the GPU cores for rendering mouse

controlled interactive object manipulations and scripted automation demos at reasonable Frames per second (FPS).

A. Tumor Interface Integrity under Gravity

Table I reports relative tumor’s centroid displacement, in terms of mean centroid error and signed interface distance (d_{max}) reporting worst interpenetration, and (d_{min}) reporting worst separation of the tumor-parenchyma interface node pairs exceeding a 1 mm separation threshold, for all three formulations at $0.1(g)$ and $1.0(g)$.

The mesh-ROI multi-material formulation preserved interface continuity at both gravity levels, with signed interface distance identically zero throughout, consistent with its shared-node construction. The constraint-coupled multi-body formulation lost continuity at both gravity levels, with 56.9% of interface pairs exceeding the separation threshold at $0.1g$ and 72.6% at $1.0(g)$; at full gravity the formulation exhibited both interpenetration (+31.6 mm) and separation (-18.4 mm) simultaneously, indicating local mesh tangling rather than uniform detachment. The uncoupled multi-body formulation remained largely stable at $0.1(g)$ (only 0.5% of pairs failed) but failed catastrophically at $1.0(g)$, with complete interface separation (100% of pairs, relative centroid displacement exceeding +250 mm).

TABLE I: Tumor Centroid Error under Gravitational Settling

FEM Model	Gravity (g)	Centroid Err. Avg. [mm]	d_{max} [mm]	d_{min} [mm]	Fail %
Uncoupled	0.1	0.052	+0.78	-1.28	0.5
Uncoupled	1.0	25.75	+248.32	-252.40	100
Coupled	0.1	0.40	+4.64	-3.75	56.9
Coupled	1.0	3.75	+31.59	-18.44	72.6
Multi-Material	0.1	0.00	0.00	0.00	0.0
Multi-Material	1.0	0.00	0.00	0.00	0.0

Since the multibody formulation is not reliable for stable palpation experiments with mesh boundary overlap and additional mesh distortion during force based interactions, we explicitly experimented our palpation hypothesis on the multi material Liver-with Tumor in different elasticities 55KPa and 75Kpa respectively, against a liver with no tumor control baseline.

B. Spatial Palpation and Stiffness interpretation

Palpation was restricted to the multi-material formulation, since Section III-A shows the multi-body formulations lose interface integrity before or during indentation and their contact forces cannot be attributed to material properties. Five fixed-depth points were indented on the multi-material model with the tumor ROI assigned 5.5, 55.0 and 75.0 kPa. Because the tumor-free liver is a separate discretization (2428 tets against 4755), it is reported for reference only. The quantitative control is the 5.5 kPa case, in which the tumor ROI is assigned the parenchymal modulus on the identical mesh, isolating the effect of multimaterial elastic assignment.

Peak reaction forces scale monotonically with tumor modulus at every contact location. Expressed as secant contact

TABLE II: Palpation response on the multi-material model. Secant contact stiffness $k = F/\delta$; apparent contrast is referenced to the 5.5 kPa homogeneous control on the same multi-material mesh. The tumor-free liver is a separate discretisation (2428 tets) and is listed for reference only.

Site	Case	δ [mm]	F [N]	k [mN/mm]	Contrast
Pt 2	No tumor*	15.14	22.5	1487.7	–
	5.5 kPa	14.74	13.8	937.3	1.000
	55 kPa	13.15	17.0	1290.6	1.377
	75 kPa	13.15	18.5	1404.8	1.499
Pt 3	No tumor*	34.90	19.2	550.7	–
	5.5 kPa	34.90	13.2	378.4	1.000
	55 kPa	34.90	15.8	452.8	1.197
	75 kPa	34.90	16.3	466.1	1.232
Pt 5	No tumor*	14.42	25.9	1798.7	–
	5.5 kPa	14.42	22.8	1579.3	1.000
	55 kPa	13.35	27.0	2022.8	1.281
	75 kPa	13.35	28.3	2120.2	1.342
Pt 6	No tumor*	28.51	24.4	855.7	–
	5.5 kPa	28.51	21.9	768.9	1.000
	55 kPa	28.51	25.5	895.3	1.164
	75 kPa	28.51	27.0	946.1	1.230
Pt 8	No tumor*	21.95	31.4	1432.6	–
	5.5 kPa	21.95	22.1	1006.5	1.000
	55 kPa	20.70	24.3	1171.6	1.164
	75 kPa	20.70	24.9	1202.6	1.195
Mean apparent contrast, 55 kPa					1.237
Mean apparent contrast, 75 kPa					1.300

TABLE III: Interactive performance by solver configuration and material contrast.

Configuration	Model	Tets	FPS
CG + uncoupled	Homogeneous liver	2428	40–50
CG + uncoupled	Multi-material, 55 kPa	4755	35
CG + uncoupled	Multi-material, 75 kPa	4755	30–32
LDL^T + generic	Multi-material, 55 kPa	4755	10–15

stiffness, the mean apparent contrast against the homogeneous control is 1.24 at 55 kPa and 1.30 at 75 kPa, ranging from 1.16 to 1.50 across sites. Points 3 and 6, where indentation depth is matched exactly between cases, give 1.20 and 1.16 at 55 kPa and 1.23 at 75 kPa. The apparent contrast is substantially below the 10-fold and 13.6-fold material contrasts, consistent with attenuation of subsurface stiffness gradients across the 1.5 cm burial depth, and indicates that surface reaction force alone resolves subsurface material contrast only weakly at clinically relevant inclusion depths.

Interactive performance degrades with both mesh size and material contrast (Table III). The conjugate gradient solver with uncoupled constraint correction sustains 40–50 FPS on the homogeneous liver, falling to 35 FPS at 55 kPa and 30–32 FPS at 75 kPa. The sparse direct solver with generic constraint correction, which retains the assembled stiffness in the compliance matrix, sustains only 10–15 FPS.

IV. DISCUSSION AND CONCLUSION

Interface continuity under gravitational loading separates the three paradigms decisively. Frictional containment alone fails under gravitational settling, and bilateral attachment constraints tear locally rather than holding the interface. Only the single mesh shared-node multi-material partition

preserves continuity identically, without a coupling operator, making this formulation reliable for building digital twins requiring stiffer inclusions within compliant substrates. Palpation observations reveal contact forces derived from Lagrange Multipliers using the uncoupled constraint correction carry limited material information (which can also be attenuated or varied by mesh tetrahedralization). However, the generic constraint correction method provides high fidelity contact data but at the cost of significant FPS drops below the threshold for interactive rehearsals. This motivates decoupling force measurement from the constraint solver with a compliant spring element between the commanded PSM jaw frame and the instrument tip to mimic a physical load cell independent of the constraint correction scheme and directly comparable to dVRK hardware. We conclude that a single volumetric mesh partitioned by regions of interest is the appropriate paradigm for interactive tissue-tumor digital twins, establishing a baseline for synthetic data generation and reinforcement-learning-based palpation policies for tumor boundary estimation.

REFERENCES

- [1] C. Huang, Q. Wang, M. Zhao, C. Chen, S. Pan, and M. Yuan, “Tactile perception technologies and their applications in minimally invasive surgery: a review,” *Frontiers in Physiology*, vol. 11, p. 611596, 2020.
- [2] C. R. Wottawa *et al.*, “Evaluating tactile feedback in robotic surgery for potential clinical application using an animal model,” *Surgical Endoscopy*, vol. 30, no. 8, pp. 3198–3209, 2016.
- [3] N. Haouchine, S. Cotin, I. Peterlik, J. Dequidt, M. S. Lopez, E. Kerrien, and M.-O. Berger, “Impact of soft tissue heterogeneity on augmented reality for liver surgery,” *IEEE transactions on visualization and computer graphics*, vol. 21, no. 5, pp. 584–597, 2014.
- [4] N. Haouchine, J. Dequidt, I. Peterlik, E. Kerrien, M.-O. Berger, and S. Cotin, “Image-guided simulation of heterogeneous tissue deformation for augmented reality during hepatic surgery,” in *2013 IEEE international symposium on mixed and augmented reality (ISMAR)*. IEEE, 2013, pp. 199–208.
- [5] J. Guo, B. Xiao, and H. Ren, “Compensating uncertainties in force sensing for robotic-assisted palpation,” *Applied Sciences*, vol. 9, no. 12, p. 2573, 2019.
- [6] A. Garg, S. Sen, R. Kapadia, Y. Jen, S. McKinley, L. Miller, and K. Goldberg, “Tumor localization using automated palpation with gaussian process adaptive sampling,” in *2016 IEEE International Conference on Automation Science and Engineering (CASE)*. IEEE, 2016, pp. 194–200.
- [7] K. A. Nichols and A. M. Okamura, “Methods to segment hard inclusions in soft tissue during autonomous robotic palpation,” *IEEE Transactions on Robotics*, vol. 31, no. 2, pp. 344–354, 2015.
- [8] K. Sangpradit, H. Liu, P. Dasgupta, K. Althoefer, and L. D. Seneviratne, “Finite-element modeling of soft tissue rolling indentation,” *IEEE Transactions on Biomedical Engineering*, vol. 58, no. 12, pp. 3319–3327, 2011.
- [9] K. Sangpradit, H. Liu, L. D. Seneviratne, and K. Althoefer, “Tissue identification using inverse finite element analysis of rolling indentation,” in *2009 IEEE International Conference on Robotics and Automation*. IEEE, 2009, pp. 1250–1255.
- [10] A. Bhattacharjee, M. T. Loghmani, and S. Anwar, “Modeling and Simulation of Robotic Palpation to Detect Subsurface Soft Tissue Anomaly for Presurgical Assessment,” *Journal of Engineering and Science in Medical Diagnostics and Therapy*, vol. 7, no. 3, p. 031002, 10 2023. [Online]. Available: <https://doi.org/10.1115/1.4063470>
- [11] R. Uppuluri, A. Bhattacharjee, S. Anwar, and Y. She, “Seebelow: Subdermal 3d reconstruction of tumors with surgical robotic palpation and tactile exploration,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024, pp. 6961–6968.
- [12] C. Duriez, C. Andriot, and A. Kheddar, “Signorini’s contact model for deformable objects in haptic simulations,” in *2004 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, vol. 4. IEEE, 2004, pp. 3232–3237.